Monopoly

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 16, Monopoly.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 16, "Monopoly".

What you must know from this chapter

1. Definition of a monopoly.
2. How can a monopoly arise?
3. Difference between P and MR.
4. Graphical representation of profits/losses.
5. Barriers to entry.
6. Comparison to perfect competition.
7. Natural monopoly.

Part 1

Monopoly and Market Power

Monopoly

Definition: Monopoly
A firm that is the sole seller of a product without close substitutes.

Definition: Market power
The ability of a firm to influence the market price of what it sells.

- A competitive firm is a price taker. A monopoly is a price maker.

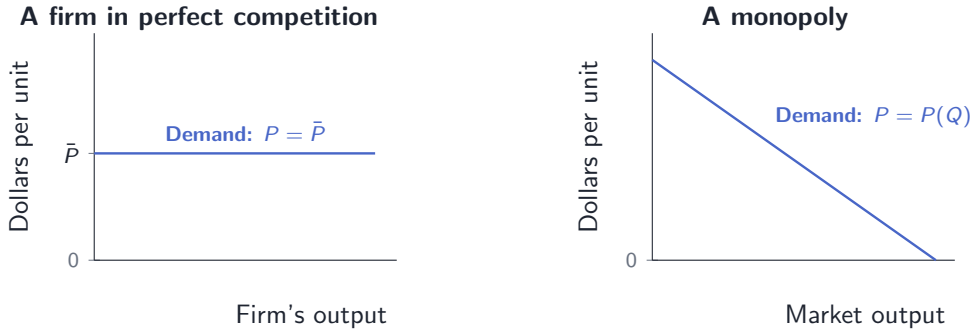
A price maker faces the market demand curve

- A competitive firm sells as many units as it likes at the market price: price is parametric, $P = \bar{P}$.
- A monopoly is the whole market. It faces the market demand curve, so its price depends on its output: $P = P(Q)$.
- Demand slopes down. If the monopoly wants to sell more, it must lower its price.
- It chooses Q; the demand curve then gives the highest price buyers will pay, $P(Q)$.

Finding

A monopoly cannot charge any price and sell any quantity. It can only choose a point on its demand curve.

Competitive firm versus monopoly



Finding

A competitive firm faces a horizontal demand line at $\bar{P}$. A monopoly faces the downward-sloping market demand curve, so selling more means a lower price.

Example: a cartridge seller

One firm is the only seller of a printer cartridge. Its demand curve:

$$P(Q) = 32 - 2Q.$$

Each extra cartridge sold lowers the price by \$2.

To sell 6 cartridges, the most it can charge is \$20. To charge \$24, it can sell only 4.

Q	$P(Q) = 32 - 2Q$
2	\$28
4	\$24
6	\$20
8	\$16

Part 1 summary

- A monopoly is the sole seller of a product without close substitutes.
- It is a price maker: it faces the market demand curve, $P = P(Q)$.
- To sell more it must lower its price. It can only choose points on its demand curve.

Your turn: Check your understanding

The cartridge seller's demand is $P(Q) = 32 - 2Q$.

1. At 4 cartridges, the price is:
   A. \$16 B. \$24 C. \$28
2. To charge \$20, it can sell:
   A. 4 B. 6 C. 10
3. A monopoly can charge any price and still sell any quantity.
   A. True B. False

Q	$P(Q)$
2	\$28
4	\$24
6	\$20
8	\$16

Solution: Check your understanding

1. **B.** $P(4) = 32 - 2 \times 4 = \$24$.
2. **B.** $20 = 32 - 2Q$, so $Q = 6$.
3. **B.** False: it can only choose points on its demand curve.

Q	$P(Q)$
2	\$28
4	\$24
6	\$20
8	\$16

Check: Part 1

Monopoly firms face

A. downward-sloping demand curves, so they can sell as much output as they desire at the market price.

B. downward-sloping demand curves, so they can sell only the specific price-quantity combinations that lie on the demand curve.

C. horizontal demand curves, so they can sell as much output as they desire at the market price.

D. horizontal demand curves, so they can sell only a limited quantity of output at each price.

Check: Part 1 – answer

Monopoly firms face

A downward-sloping demand curves, so they can sell as much output as they desire at the market price.

B downward-sloping demand curves, so they can sell only the specific price-quantity combinations that lie on the demand curve.

C horizontal demand curves, so they can sell as much output as they desire at the market price.

D horizontal demand curves, so they can sell only a limited quantity of output at each price.

Finding

A monopoly is the whole market, so it faces the downward-sloping market demand curve. Each output has one highest price, $P(Q)$. Horizontal demand describes a competitive firm.

Part 2

Barriers to Entry and Natural Monopoly

Barriers to entry

Definition: Barrier to entry

Anything that stops other firms entering a market and competing.

Without entry barriers, profit attracts entry and the monopoly disappears. Monopolies arise from three sources:

1. **A key resource** is owned by one firm (a unique mineral deposit).
2. **The government** gives one firm the exclusive right to sell (a patent or a copyright).
3. **Costs:** one firm can supply the whole market more cheaply than two or more firms. This is a natural monopoly.

Natural monopoly

Definition: Natural monopoly

A market in which one firm can supply the whole market at a lower average total cost than two or more firms could.

Example: the water supply network in a town. Laying the pipes is expensive. Once they are laid, connecting one more home costs little. One network serves the whole town more cheaply than two networks side by side.

Why one firm is cheaper: falling average total cost

- A natural monopoly arises when fixed cost FC is large and marginal cost c is small and constant: $\text{TC} = \text{FC} + c \times Q$.
- Then $\text{ATC} = \dfrac{\text{FC}}{Q} + c$. As Q rises, FC is spread over more units, so ATC falls.

Finding

If ATC falls over the whole market, splitting output between two firms raises ATC: each firm pays the fixed cost.

Example: a town water network

Laying the pipes costs \$120 (FC). Each water connection to a home costs \$3 (MC):

$$\text{TC} = 120 + 3Q, \qquad \text{ATC} = \frac{120}{Q} + 3.$$

ATC falls as water connections rise: \$9 at 20, \$5 at 60.

ATC is always above MC, which is \$3.

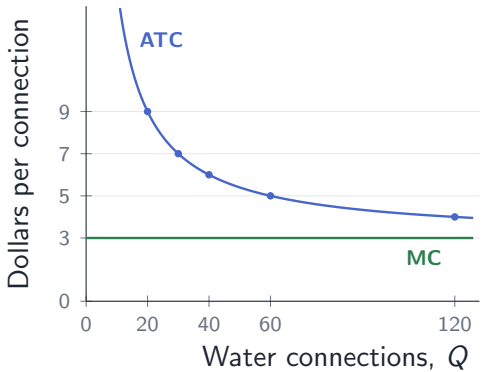
Q	TC	ATC
20	\$180	\$9
30	\$210	\$7
40	\$240	\$6
60	\$300	\$5

Natural monopoly on the graph

Q is the number of water connections to homes. ATC $= 120/Q + 3$ falls over the whole range.

MC is flat at \$3, below ATC at every output.

The more homes one water network connects, the lower its cost per connection.



One network or two?

The town needs 60 water connections.

- One network: $\text{TC} = 120 + 3 \times 60 = \$300$. $\text{ATC} = \$5$.
- Two water networks of 30 each: $2 \times (120 + 3 \times 30) = 2 \times 210 = \$420$. $\text{ATC} = \$7$.
- Two networks cost $420 - 300 = \$120$ more: the second fixed cost.

Finding

In a natural monopoly, one firm supplies the market at the lowest cost. A second firm would duplicate the fixed cost.

Part 2 summary

- Barriers to entry are the cause of monopoly: without them, entry competes profit away.
- Three sources: a key resource, government-created rights, and costs.
- Natural monopoly: large FC and low MC make ATC fall as output rises, so one firm supplies the market most cheaply.

Your turn: Check your understanding

The town water network: $\text{TC} = 120 + 3Q$.

1. ATC at 40 connections is:

   A. \$3 B. \$6 C. \$240

2. Serving 40 connections with two networks of 20 instead of one costs extra:

   A. \$0 B. \$60 C. \$120

3. A patent creates a barrier to entry.

   A. True B. False

Q	TC	ATC
20	\$180	\$9
40	\$240	\$6

Solution: Check your understanding

1. **B.** ATC $= 120/40 + 3 = 3 + 3 = \$6$.
2. **C.** Two networks: $2 \times 180 = \$360$; one: \$240. Extra: $360 - 240 = \$120$.
3. **A.** True: the government gives one firm the exclusive right to sell.

Q	TC	ATC
20	\$180	\$9
40	\$240	\$6

Check: Part 2

When an industry is a natural monopoly,

A it is characterized by constant returns to scale.

B it is characterized by diseconomies of scale.

C a larger number of firms may lead to a lower average total cost.

D a larger number of firms will lead to a higher average total cost.

Check: Part 2 – answer

When an industry is a natural monopoly,

A it is characterized by constant returns to scale.

B it is characterized by diseconomies of scale.

C a larger number of firms may lead to a lower average total cost.

D a larger number of firms will lead to a higher average total cost.

Finding

ATC falls as one firm's output rises. Splitting output among more firms means each pays the fixed cost and produces less, so ATC is higher: the network's \$5 becomes \$7 with two firms.

Part 3

Total, Average and Marginal Revenue

Total, average and marginal revenue

Definition: Total revenue

$\text{TR}(Q) = P(Q) \times Q$: the price depends on output.

Definition: Average revenue

$\text{AR} = \text{TR}/Q = P(Q)$: revenue per unit is the price, as in competition.

Definition: Marginal revenue

$\text{MR} = \Delta\text{TR}/\Delta Q$: the extra revenue from an extra unit. $\Delta Q = 1$.

Selling more changes revenue in two ways

Total revenue is $\text{TR} = P \times Q$. Raise output by ΔQ. To sell the extra units, the price must change by ΔP (a fall, so $\Delta P < 0$).

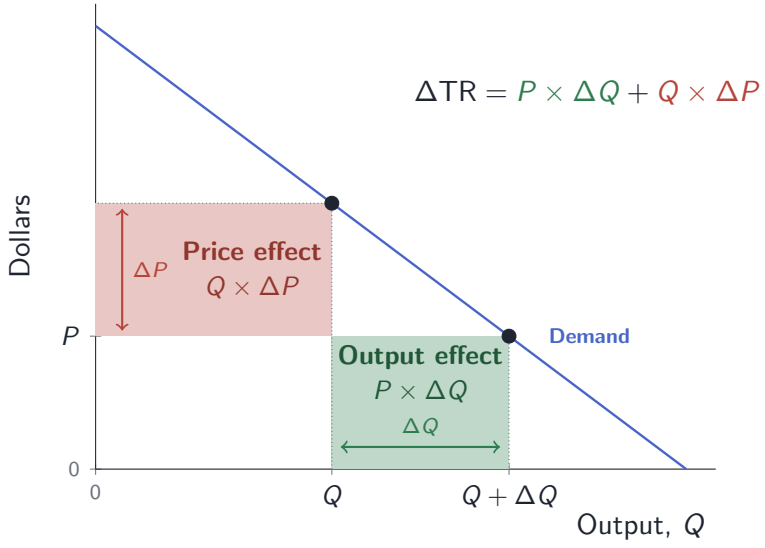
- **Output effect:** the extra ΔQ units sell at price P: revenue rises by $P \times \Delta Q$.
- **Price effect:** the Q units already sold now sell at a lower price: revenue changes by $Q \times \Delta P$, which is negative.
- Together:

$$\Delta \text{TR} = P \times \Delta Q + Q \times \Delta P.$$

Finding

P is the new price; Q is the number of units already sold.

The fundamental trade-off



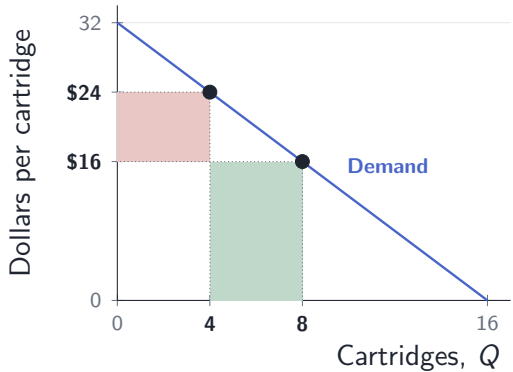
The two effects on the graph

The cartridge seller goes from 4 to 8 units. The price falls from \$24 to \$16: $\Delta Q = 4$, $\Delta P = -8$.

Output effect (green): 4 more units at \$16: $16 \times 4 = +\$64$.

Price effect (red): the first 4 units lose \$8 each: $4 \times (-8) = -\$32$.

$\Delta \text{TR} = 64 - 32 = \$32$. Check: TR goes from \$96 to \$128.



From the change in revenue to marginal revenue

- Start from the two effects:

$$\Delta \text{TR} = P \times \Delta Q + Q \times \Delta P.$$

- Divide both sides by ΔQ:

$$\frac{\Delta \text{TR}}{\Delta Q} = P \times \frac{\Delta Q}{\Delta Q} + Q \times \frac{\Delta P}{\Delta Q}.$$

- $\Delta Q / \Delta Q = 1$ and $\Delta \text{TR} / \Delta Q = \text{MR}$, so

$$\text{MR} = P + Q \times \frac{\Delta P}{\Delta Q}.$$

Marginal revenue is below price (= average revenue)

$$\text{MR} = P + Q \times \frac{\Delta P}{\Delta Q}$$

- Demand slopes down, so $\Delta P/\Delta Q < 0$. So the last term is negative.
- Write it as $-k$, where $k > 0$. Then $\text{MR} = P - k$, so

$$P = \text{MR} + k.$$

- Whatever MR is, P is bigger by k. So $P > \text{MR}$.
- $\text{AR} = P$, so also $\text{AR} > \text{MR}$.

MR for a straight-line demand curve

Demand is a straight line: $P = a - bQ$, with $a > 0$ and $b > 0$. Each extra unit lowers the price by b, so $\Delta P / \Delta Q = -b$.

$$
\begin{aligned}
\text{MR} &= P + Q \times \frac{\Delta P}{\Delta Q} \\
&= (a - bQ) + Q \times \frac{\Delta P}{\Delta Q} && \text{replace } P \text{ with } a - bQ \\
&= (a - bQ) + Q \times (-b) && \text{replace } \Delta P / \Delta Q \text{ with } -b \\
&= a - bQ - bQ && Q \times (-b) = -bQ \\
&= a - 2bQ && -bQ - bQ = -2bQ
\end{aligned}
$$

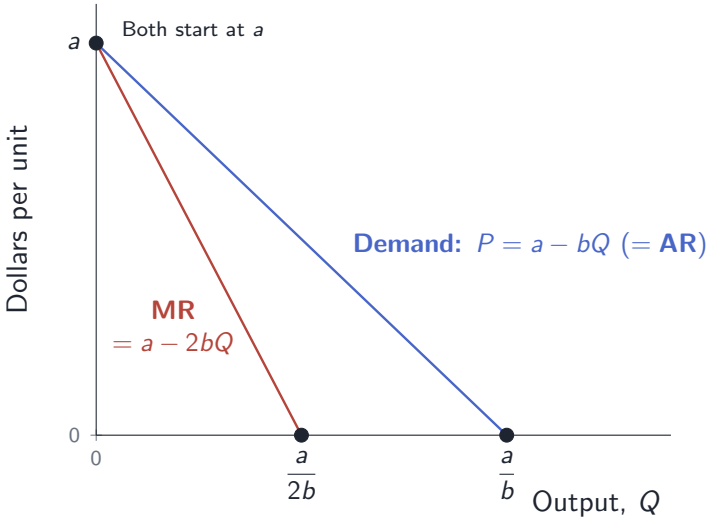
MR is twice as steep as demand

- Demand: $P = a - bQ$. Each extra unit lowers the price by b.
- Marginal revenue: $\text{MR} = a - 2bQ$. Each extra unit lowers MR by $2b$.
- Both lines start at a on the vertical axis.

Finding

For every straight-line demand curve, the MR line starts at the same point and is twice as steep. This is a general result, not a feature of one example.

Demand (AR) and marginal revenue



Example: the cartridge seller's revenue

Q	$P(Q)$	TR	MR
1	\$30	\$30	\$30
2	\$28	\$56	\$26
3	\$26	\$78	\$22
4	\$24	\$96	\$18
5	\$22	\$110	\$14
6	\$20	\$120	\$10
7	\$18	\$126	\$6
8	\$16	\$128	\$2
9	\$14	\$126	−\$2

$P(Q) = 32 - 2Q$; $\text{TR} = P(Q) \times Q$; MR in row Q is $\text{TR}(Q) - \text{TR}(Q-1)$.

MR is below the price for every unit after the first.

The 9th unit has negative MR: TR falls from \$128 to \$126.

The price falls \$2 per unit; MR falls \$4 per unit: twice as steep.

The two effects in the table

Cartridges: $\Delta P/\Delta Q = -2$. Use $\text{MR} = P + Q \times \Delta P/\Delta Q$, with P the new price and Q the units already sold.

- **6th unit:** $\text{MR} = 20 + 5 \times (-2) = 20 - 10 = \$10$.
- **9th unit:** $\text{MR} = 14 + 8 \times (-2) = 14 - 16 = -\$2$.

Finding

When the price effect is larger than the output effect, MR is negative: selling more lowers total revenue.

Part 3 summary: revenue

- Total revenue: $\text{TR}(Q) = P(Q) \times Q$.
- Average revenue equals price: $\text{AR} = P(Q)$.
- Selling more changes revenue in two ways: $\Delta\text{TR} = P \times \Delta Q + Q \times \Delta P$.
- The output effect, $P \times \Delta Q$, raises revenue.
- The price effect, $Q \times \Delta P$, lowers revenue.
- Marginal revenue: $\text{MR} = P + Q \times \Delta P / \Delta Q$.

Part 3 summary: marginal revenue

- For a monopoly, demand slopes down, so $P > \text{MR}$.
- Because $\text{AR} = P$, also $\text{AR} > \text{MR}$.
- For a straight-line demand curve, $\text{MR} = a - 2bQ$: it starts at the same point and is twice as steep.
- MR can be negative. Then selling more lowers total revenue.
- Compared with competition: a competitive firm's price does not change when it sells more, so $\text{MR} = \bar{P}$.

Your turn: Check your understanding

The cartridge seller, $P(Q) = 32 - 2Q$.

1. TR at 4 cartridges is:
   A. \$24 B. \$78 C. \$96
2. MR of the 4th cartridge is:
   A. \$18 B. \$24 C. \$96
3. For the 4th cartridge, the price effect changes revenue by:
   A. $-\$2$ B. $-\$6$ C. $-\$24$

Q	$P(Q)$	TR
3	\$26	\$78
4	\$24	\$96

Solution: Check your understanding

1. **C.** TR $= 24 \times 4 = \$96$.
2. **A.** MR $= 96 - 78 = \$18$.
3. **B.** $Q \times \Delta P = 3 \times (-2) = -\$6$. Check: $24 - 6 = \$18$.

Q	$P(Q)$	TR
3	\$26	\$78
4	\$24	\$96

Check: Part 3

If a profit-maximizing monopolist faces a downward-sloping market demand curve, its

A average revenue is less than the price of the product.

B average revenue is less than marginal revenue.

C marginal revenue is less than the price of the product.

D marginal revenue is greater than the price of the product.

Check: Part 3 – answer

If a profit-maximizing monopolist faces a downward-sloping market demand curve, its

A average revenue is less than the price of the product.

B average revenue is less than marginal revenue.

C marginal revenue is less than the price of the product.

D marginal revenue is greater than the price of the product.

Finding

To sell one more unit, the monopoly lowers the price on all its units. The price effect makes MR $< P$. Average revenue always equals the price, so A and B are wrong.

Part 4

Profit Maximization and Profit or Loss

The rule carries over from competition

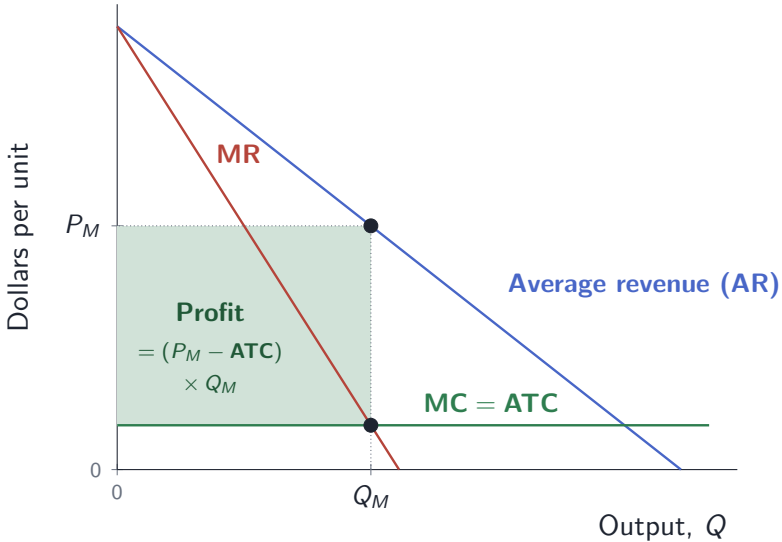
- Profit depends on output: $\pi(Q) = \text{TR}(Q) - \text{TC}(Q)$.
- One more unit changes profit by $\Delta\pi = \text{MR} - \text{MC}$.
- If $\text{MR} > \text{MC}$, increase output. If $\text{MC} > \text{MR}$, cut output.
- Profit is maximized where $\text{MR} = \text{MC}$.

Finding

The profit-maximizing rule is $\text{MR} = \text{MC}$. For one more unit:

$$\text{MR} = \text{MC} \;\Rightarrow\; \text{MR} - \text{MC} = 0 \;\Rightarrow\; \Delta\pi = \Delta\text{TR} - \Delta\text{TC} = 0.$$

Optimal output, price and profit



Example: the cartridge seller's optimal output

Q	$P(Q)$	TR	MR	MC	Profit
4	\$24	\$96	\$18	\$10	\$56
5	\$22	\$110	\$14	\$10	\$60
6	\$20	\$120	\$10	\$10	\$60
7	\$18	\$126	\$6	\$10	\$56
8	\$16	\$128	\$2	\$10	\$48

Each cartridge costs \$10 to make. No fixed cost, so ATC = MC = \$10 and profit = TR − $10Q$.

Unit 6: MR = MC = \$10, so $\Delta\pi = 0$. Optimal output 6.

Price from demand: $P(6) = \$20$, above MC = \$10.

Profit = $(20 - 10) \times 6 = \$60$.

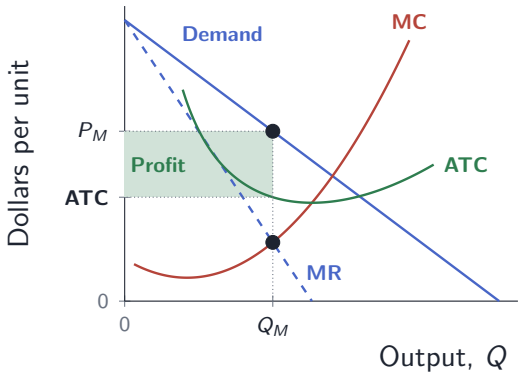
Source of the numbers: $P = 32 - 2Q$, so TR = $(32 - 2Q) \times Q$; TC = $10Q$; profit = TR − TC.

Profit on the graph

A general monopoly with U-shaped costs.

Choose Q_M where MR = MC. Read P_M from demand and ATC at Q_M.

$P_M >$ ATC: profit. Rectangle height $P_M -$ ATC, width Q_M.



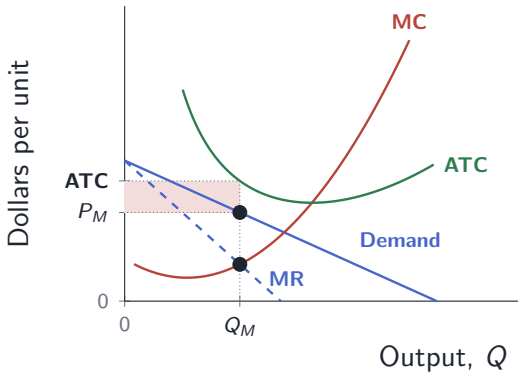
Loss on the graph

The same costs, weaker demand.

Q_M where MR = MC; P_M from demand.

$P_M <$ ATC: a loss. Rectangle height ATC $- P_M$, width Q_M.

Being a monopoly does not guarantee a profit.



A monopoly has no supply curve

- A supply curve gives the quantity a firm offers at each price it takes as given.
- A monopoly does not take the price as given. It chooses output and price together from MR, MC and demand.
- The same MC can go with different prices, depending on the shape of demand.

Finding

A competitive firm's supply curve is its MC curve above minimum AVC. A monopoly has no supply curve: its output depends on demand.

Price-cost margins and the elasticity of demand

- Profit is revenue minus cost: $\pi = \text{TR} - \text{TC}$.
- One more unit changes profit by

$$\frac{\Delta\pi}{\Delta Q} = \frac{\Delta\text{TR}}{\Delta Q} - \frac{\Delta\text{TC}}{\Delta Q}.$$

- $\Delta\text{TR}/\Delta Q = \text{MR}$ and $\Delta\text{TC}/\Delta Q = \text{MC}$. Write c for MC.
- Profit is maximized where $\Delta\pi/\Delta Q = 0$, so $\text{MR} = c$.

Write out marginal revenue

- From Part 3, $\text{MR} = P + Q \times \dfrac{\Delta P}{\Delta Q}$.
- So $\text{MR} = c$ becomes
$$P + Q \times \frac{\Delta P}{\Delta Q} = c.$$
- Subtract c, and move the last term to the right-hand side:
$$P - c = -Q \times \frac{\Delta P}{\Delta Q}.$$
- Divide both sides by P:
$$\frac{P - c}{P} = -\frac{Q}{P} \times \frac{\Delta P}{\Delta Q}.$$

The right-hand side is one over the elasticity

- The price elasticity of demand (Chapter 5):

$$\varepsilon = \frac{\%\Delta Q}{\%\Delta P} = \frac{\Delta Q/Q}{\Delta P/P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}.$$

- Demand slopes down, so $\varepsilon < 0$. Its size is $|\varepsilon| = -\varepsilon$.
- Turn it upside down: $\dfrac{1}{\varepsilon} = \dfrac{\Delta P}{\Delta Q} \times \dfrac{Q}{P}$.
- So the right-hand side is $-\dfrac{1}{\varepsilon} = \dfrac{1}{|\varepsilon|}$.

Finding

The profit-maximizing monopoly sets $\dfrac{P-c}{P} = \dfrac{1}{|\varepsilon|}$: the price-cost margin is one over the elasticity of demand.

Elastic demand: $|\varepsilon| > 1$

$$\frac{P - c}{P} = \frac{1}{|\varepsilon|}$$

- If $|\varepsilon| > 1$, the margin $1/|\varepsilon|$ is between 0 and 1.
- $|\varepsilon| = 2$: the margin is $\frac{1}{2}$. So $P - c = \frac{1}{2}P$, and $P = 2c$.
- $|\varepsilon| = 5$: the margin is $\frac{1}{5}$. So $P - c = \frac{1}{5}P$, and $P = 1.25c$.

Finding

The more elastic is demand, the smaller is the margin of price over marginal cost.

Unit elastic and inelastic demand

$$\frac{P - c}{P} = \frac{1}{|\varepsilon|}$$

- **Unit elastic,** $|\varepsilon| = 1$: the margin is 1. So $P - c = P$, and $c = 0$. Here MR $= 0$.
- **Inelastic,** $|\varepsilon| < 1$: the margin is above 1. So $P - c > P$, and $c < 0$.
- Marginal cost cannot be negative. So the rule never holds where demand is inelastic.

Finding

With $c > 0$, a profit-maximizing monopoly always produces on the elastic part of the demand curve, where $|\varepsilon| > 1$.

Perfectly elastic demand: $|\varepsilon| \to \infty$

- As $|\varepsilon|$ grows without limit, $1/|\varepsilon|$ goes to 0.
- So $(P - c)/P = 0$, and $P = c$.
- This is the competitive firm: demand is horizontal at the market price, and $P = \text{MC}$.

Demand	$\lvert\varepsilon\rvert$	Margin $(P - c)/P$
Perfectly elastic	∞	0
Elastic	5	$\frac{1}{5}$
Elastic	2	$\frac{1}{2}$
Unit elastic	1	1 (only if $c = 0$)
Inelastic	below 1	not possible

Part 4 summary

- Profit is maximized where MR = MC: $\Delta\pi = \text{MR} - \text{MC} = 0$.
- The price comes from the demand curve at that output: $P_M = P(Q_M)$.
- Because $P >$ MR, the monopoly price is above marginal cost.
- Profit $= (P_M - \text{ATC}) \times Q_M$. If $P_M <$ ATC, the monopoly makes a loss.
- A monopoly has no supply curve.
- The price-cost margin is $(P - c)/P = 1/|\varepsilon|$. A monopoly produces where demand is elastic.

Your turn: Check your understanding

The cost of making each extra cartridge (marginal cost) rises to \$14. There is no fixed cost.

1. The optimal (best) output is:
   A. 3 B. 5 C. 9
2. The price is:
   A. \$14 B. \$22 C. \$24
3. The profit is:
   A. \$40 B. \$70 C. \$110

Q	$P(Q)$	TR	MR
4	\$24	\$96	\$18
5	\$22	\$110	\$14
6	\$20	\$120	\$10

Solution: Check your understanding

1. **B.** Unit 5: MR $= \$14 =$ MC.
2. **B.** From demand: $P(5) = 32 - 10 = \$22$.
3. **A.** Profit $= (22 - 14) \times 5 = \$40$.

Q	$P(Q)$	TR	MR
4	\$24	\$96	\$18
5	\$22	\$110	\$14
6	\$20	\$120	\$10

Check: Part 4

Kayla's Seitan Shop is the only place within 100 miles that sells seitan Buffalo-style wings. Assuming that Kayla is a monopolist and maximizing her profit, which of the following statements is true?

A The price of Kayla's seitan wings will be less than Kayla's marginal cost.

B The price of Kayla's seitan wings will exceed Kayla's marginal cost.

C The price of Kayla's seitan wings will equal Kayla's marginal cost.

D Costs are irrelevant to Kayla because she is a monopolist.

Check: Part 4 – answer

A profit-maximizing monopolist: which statement is true?

A The price of Kayla's seitan wings will be less than Kayla's marginal cost.

B The price of Kayla's seitan wings will exceed Kayla's marginal cost.

C The price of Kayla's seitan wings will equal Kayla's marginal cost.

D Costs are irrelevant to Kayla because she is a monopolist.

Finding

Kayla produces where MR $=$ MC and charges the price on her demand curve. For a monopoly $P >$ MR, so $P >$ MC. Costs matter: MC sets her output.

Part 5

Monopoly versus Perfect Competition

Two output rules

- Both firms produce where MR $=$ MC.
- Competition: MR $= \bar{P}$, so $P =$ MC.
- Monopoly: MR $< P$, so $P >$ MC.
- With the same demand and costs, the monopoly stops where MR, not price, meets MC: a smaller output at a higher price.

Finding

Monopoly: lower quantity, higher price. Competition: output where $P =$ MC.

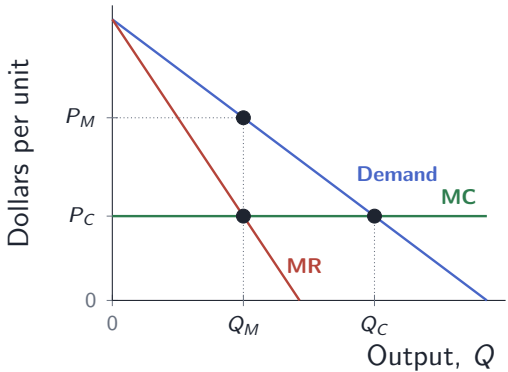
Monopoly and competition on one graph

Same demand and constant MC.

Competition: Q_C where demand meets MC; price $P_C = \text{MC}$.

Monopoly: Q_M where MR meets MC; price P_M from demand.

$Q_M < Q_C$ and $P_M > P_C$.



Example: printer cartridges – the set-up

Consider an example. A firm is the only seller of a printer cartridge.

- The inverse demand function is $P = 32 - 2Q$.
- Each extra cartridge sold lowers the price by \$2.
- Total revenue is price times quantity: $\text{TR} = P \times Q$.
- Costs are a simple straight line: $\text{TC} = c \times Q$, with $c = \$10$.
- Each cartridge costs \$10 to make, so the marginal cost is $\text{MC} = \$10$.

Finding

Demand: $P = 32 - 2Q$. Revenue: $\text{TR} = P \times Q$. Cost: $\text{TC} = 10Q$, so $\text{MC} = \$10$.

Example: the monopoly output and price

The monopoly sells one more cartridge as long as it adds at least as much to revenue as to cost: MR $\geq$ MC = \$10.

- **5th cartridge:** TR rises from $4 \times \$24 = \$96$ to $5 \times \$22 = \$110$. MR = \$14 > \$10. Sell it.
- **6th cartridge:** TR rises from \$110 to $6 \times \$20 = \$120$. MR = \$10 = MC. Sell it.
- **7th cartridge:** TR rises from \$120 to $7 \times \$18 = \$126$. MR = \$6 < \$10. Do not sell it.
- So MR = MC at $Q_M = 6$. The price comes from demand: $P_M = 32 - 2 \times 6 = \$20$.

Example: the competitive output and price

- Under competition, output expands until price equals marginal cost: $P = \text{MC}$.
- Set demand equal to MC: $32 - 2Q = 10$.
- Add $2Q$ to both sides and subtract 10: $22 = 2Q$, so $Q_C = 11$.
- The price is $P_C = \text{MC} = \$10$.

	Q	P	$P - \text{MC}$
Monopoly	6	\$20	\$10
Competition	11	\$10	\$0

Finding

The monopoly sells 5 fewer cartridges, at a price \$10 higher.

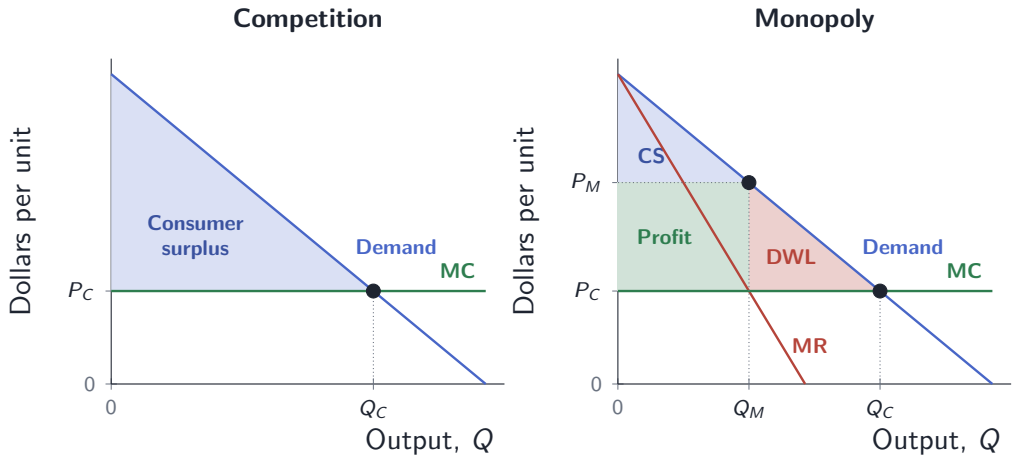
The deadweight loss of monopoly

Definition: Deadweight loss

The fall in total surplus when output is below the efficient level.

- The efficient output is where $P = \text{MC}$: buyers' value of the last unit equals its cost.
- Competition produces there, at Q_C.
- The monopoly stops earlier, at Q_M, where $\text{MR} = \text{MC}$.
- For each unit between Q_M and Q_C, buyers value it above its cost, but it is not made. That lost surplus is the deadweight loss.

Surplus under competition and under monopoly



The deadweight loss: findings

- Competition: price $P_C = \text{MC}$, output Q_C. All the surplus goes to buyers as consumer surplus.
- Monopoly: price rises to P_M and output falls to Q_M.
- Consumer surplus shrinks to the small triangle above P_M.
- Part of the lost consumer surplus becomes profit, $(P_M - \text{MC}) \times Q_M$. It moves from buyers to the firm.
- The rest, the triangle from Q_M to Q_C, goes to no one. That is the deadweight loss.

Finding

Monopoly profit is a transfer from buyers to the firm. The deadweight loss is surplus that no one gets: the social cost of monopoly.

The two market structures

	Perfect competition	Monopoly
Number of firms	Many	One
Product	Identical	No close substitutes
Entry	Free	Blocked by barriers
Demand facing the firm	Horizontal at $\bar{P}$	Market demand, $P(Q)$
Price and MR	$P = \text{MR}$	$P > \text{MR}$
Output rule	$P = \text{MC}$	$\text{MR} = \text{MC}$, $P > \text{MC}$
Long-run profit	Zero	Can stay positive
Deadweight loss	None	Yes

Part 5 summary

- Competition produces where $P = \text{MC}$; monopoly where $\text{MR} = \text{MC}$, with $P > \text{MC}$.
- With the same demand and costs, monopoly output is lower and its price higher.
- The units between Q_M and Q_C are worth more to buyers than they cost: their lost surplus is the deadweight loss, $\frac{1}{2} \times (Q_C - Q_M) \times (P_M - \text{MC})$ for straight-line demand and constant MC.
- Monopoly profit is a transfer, not a deadweight loss.

Your turn: Check your understanding

Cartridges: $P(Q) = 32 - 2Q$, cost \$14 a cartridge.
The monopoly sells 5 at \$22 (Part 4).

1. The competitive output is:
   A. 5 B. 9 C. 14
2. The monopoly price exceeds MC by:
   A. \$0 B. \$8 C. \$22
3. The deadweight-loss triangle is:
   A. \$16 B. \$32 C. \$40

	Q	P
Monopoly	5	\$22
Competition	?	\$14

Solution: Check your understanding

1. **B.** $P = \text{MC}$: $32 - 2Q = 14$, so $Q = 9$.
2. **B.** $22 - 14 = \$8$.
3. **A.** Triangle:
   $\frac{1}{2} \times (9 - 5) \times (22 - 14) = \frac{1}{2} \times 4 \times 8 = \$16$.

	Q	P
Monopoly	5	\$22
Competition	9	\$14

Check: Part 5

Monopolies are socially inefficient because the price they charge is

A equal to marginal revenue.

B above marginal cost.

C equal to demand.

D above demand.

Check: Part 5 – answer

Monopolies are socially inefficient because the price they charge is

A equal to marginal revenue.

B above marginal cost.

C equal to demand.

D above demand.

Finding

With $P >$ MC, some buyers value extra units above their cost, but the monopoly does not produce them. That lost surplus is the deadweight loss. Efficiency needs $P =$ MC.

Part 6

Price Discrimination and Public Policy

What is price discrimination?

Definition: Price discrimination

Selling the same good at different prices to different buyers, when the difference in price is not due to a difference in cost.

Examples

- **Airline seats:** two passengers on the same flight pay very different fares, depending on when they book.
- **Cinema tickets:** people over 65 pay less for the same film on a Thursday afternoon.

What a firm needs to price discriminate

1. **Market power.** A competitive firm takes the price as given, so it cannot set different prices.
2. **A way to segment the market.** The firm must sort buyers into groups that will pay different amounts (by booking date, by age).
3. **No resale.** Buyers who pay the low price must not be able to resell to buyers who would pay the high price.

Finding

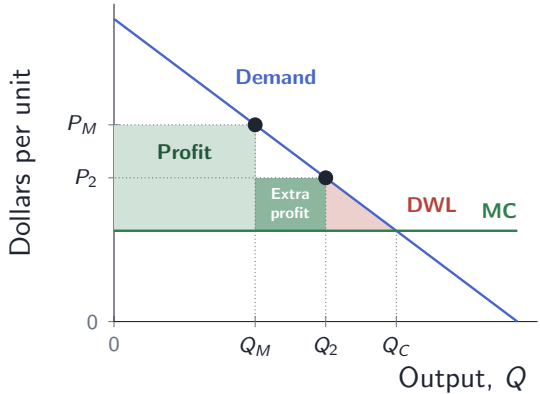
The airline can do all three: it is one of few sellers on a route, booking date sorts travelers, and tickets carry the passenger's name.

Two prices instead of one

With one price, the monopoly sells Q_M at P_M (Part 4). Units from Q_M to Q_C are lost: the deadweight loss.

Now add a lower price, P_2, only for buyers who will not pay P_M.

Output rises to Q_2. Profit rises by the dark rectangle. The deadweight loss shrinks.



Finding

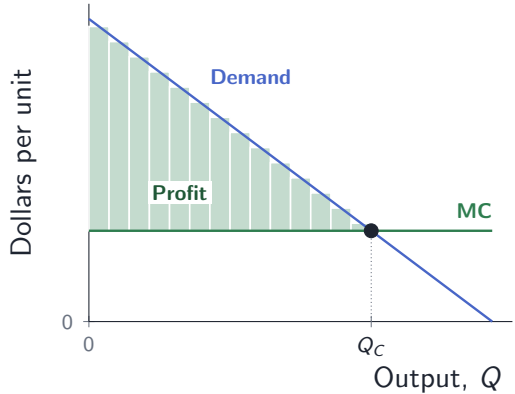
Price discrimination raises profit and output, and cuts the deadweight loss.

Perfect price discrimination

Now charge every buyer exactly the most that buyer will pay. Each strip is one buyer.

One more sale does not lower the price to anyone else, so MR $= P$.

The firm sells until $P =$ MC: output Q_C, the competitive output.



Finding

No deadweight loss: every unit worth more than MC is sold. All surplus is profit.

Public policy toward monopolies

1. **Antitrust law:** block mergers and break up firms that reduce competition.
2. **Regulation:** set the price the monopoly may charge.
3. **Public ownership:** the government runs the monopoly.
4. **Do nothing:** when the cure costs more than the deadweight loss.

Finding

Natural monopoly is the hard case: breaking it up would raise costs, so the usual choice is regulation or public ownership.

Regulating a natural monopoly

The town water network: $\text{TC} = 120 + 3Q$, $\text{MC} = \$3$. At 60 connections, $\text{ATC} = \$5$.

- **Price = MC = \$3:** efficient, but $(5 - 3) \times 60 = \$120$ loss, the whole fixed cost. The firm needs a subsidy or it leaves.
- **Price = ATC = \$5:** revenue $5 \times 60 = \$300$ covers $\text{TC} = \$300$: zero profit. But $P > \text{MC}$: homes take fewer water connections at \$5 than at \$3, so some deadweight loss remains.

Finding

For a natural monopoly ATC is above MC, so a price equal to MC always makes a loss.

Part 6 summary

- Price discrimination: the same good at different prices to different buyers.
- It needs market power, a way to segment the market, and no resale.
- A second, lower price raises profit and output, and cuts the deadweight loss.
- Perfect price discrimination: output Q_C, no deadweight loss, all surplus to the firm.
- Policy options: antitrust, regulation, public ownership, or doing nothing.
- Regulating a natural monopoly at $P = \text{MC}$ causes a loss; at $P = \text{ATC}$ it breaks even.

Your turn: Check your understanding

1. Which is price discrimination?
   A. A larger TV costs more than a smaller one.
   B. A cinema charges people over 65 less for the same film.
   C. A store raises its price after its costs rise.
2. An airline can charge two fares on one flight only if:
   A. tickets cannot be resold. B. it has many rivals. C. every seat costs the same.
3. With perfect price discrimination, the deadweight loss is:
   A. larger than with one price. B. zero. C. unchanged.

Solution: Check your understanding

1. **B.** Same film, same cost, different prices. In A the goods differ; in C the price follows cost.
2. **A.** If a cheap ticket could be resold, the high-fare buyers would buy it.
3. **B.** Every unit worth more than MC is sold: output reaches Q_C.

Check: Part 6

Price discrimination

A is illegal in the United States and Europe.

B can occur in both perfectly competitive and monopoly markets.

C is illogical because it does not maximize profits.

D can maximize profits if the seller can prevent the resale of goods between customers.

Check: Part 6 – answer

Price discrimination

A is illegal in the United States and Europe.

B can occur in both perfectly competitive and monopoly markets.

C is illogical because it does not maximize profits.

D can maximize profits if the seller can prevent the resale of goods between customers.

Finding

Charging each group what it will pay raises profit, as on the two-price graph. It needs market power (so not competition) and no resale between buyers.

Which rule answers which question

Question	Rule
What demand does a monopoly face?	The market demand curve: $P = P(Q)$.
Why does a monopoly exist?	Barriers to entry: a key resource, government rights, or natural monopoly.
How do P and MR compare?	MR $< P$: the price effect.
How much should it produce?	MR $=$ MC. Increase output if MR $>$ MC; cut it if MC $>$ MR.
What price does it charge?	The price on the demand curve at that output: $P >$ MC.
Does it make a profit?	$\pi = (P - \text{ATC}) \times Q$; a loss if $P <$ ATC.
How does it compare with competition?	Lower output, higher price, a deadweight loss.